

Inverse Functions

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: 2 # _____ The point $(3, 7)$ lies on the graph of f. State the corresponding point on f^{-1}.	ANSWER: no # _____ Find the inverse of $f(x) = x^3 + 4$.
ANSWER: $\frac{x-3}{2}$ # _____ The function f satisfies $f(2) = 3$, $f(4) = 5$, $f(5) = 9$. Find $f^{-1}(5)$.	ANSWER: 4 # _____ Is $f(x) = x^2$ (on all real numbers) one-to-one? Answer yes or no.

ANSWER: $5x - 1$ # _____ Find the inverse of $f(x) = 2x + 3$.	ANSWER: (b, a) # _____ Find the inverse of $f(x) = 7x$.
ANSWER: $\sqrt[3]{x - 4}$ # _____ The point (a, b) lies on the graph of f. State the corresponding point on f^{-1}.	ANSWER: $(7, 3)$ # _____ Find the inverse of $f(x) = x^3 - 4$.
ANSWER: $\sqrt[3]{x + 4}$ # _____ Find the inverse of $f(x) = \frac{x + 1}{5}$.	ANSWER: $\frac{x}{7}$ # _____ Given $f(x) = x^3$, find $f^{-1}(8)$.